

Module -1

AIR POLLUTION



introduction

- Air is essential for life it self, without it we could survive only a few minutes.
- It constitutes immediate physical environment of living organisms
- The atmosphere is layered in to four distinct which are: Troposphere, stratosphere, mesosphere, and thermosphere.

THE EARTH'S ATMOSPHERE

EXOSPHERE

>700 TO 190,000 KM

EXOBASE

>700 TO 1,000 KM

THERMOSPHERE

80 TO 700 KM

KARMAN LINE

100 KM

MESOSPHERE

50 TO 80 KM

STRATOSPHERE

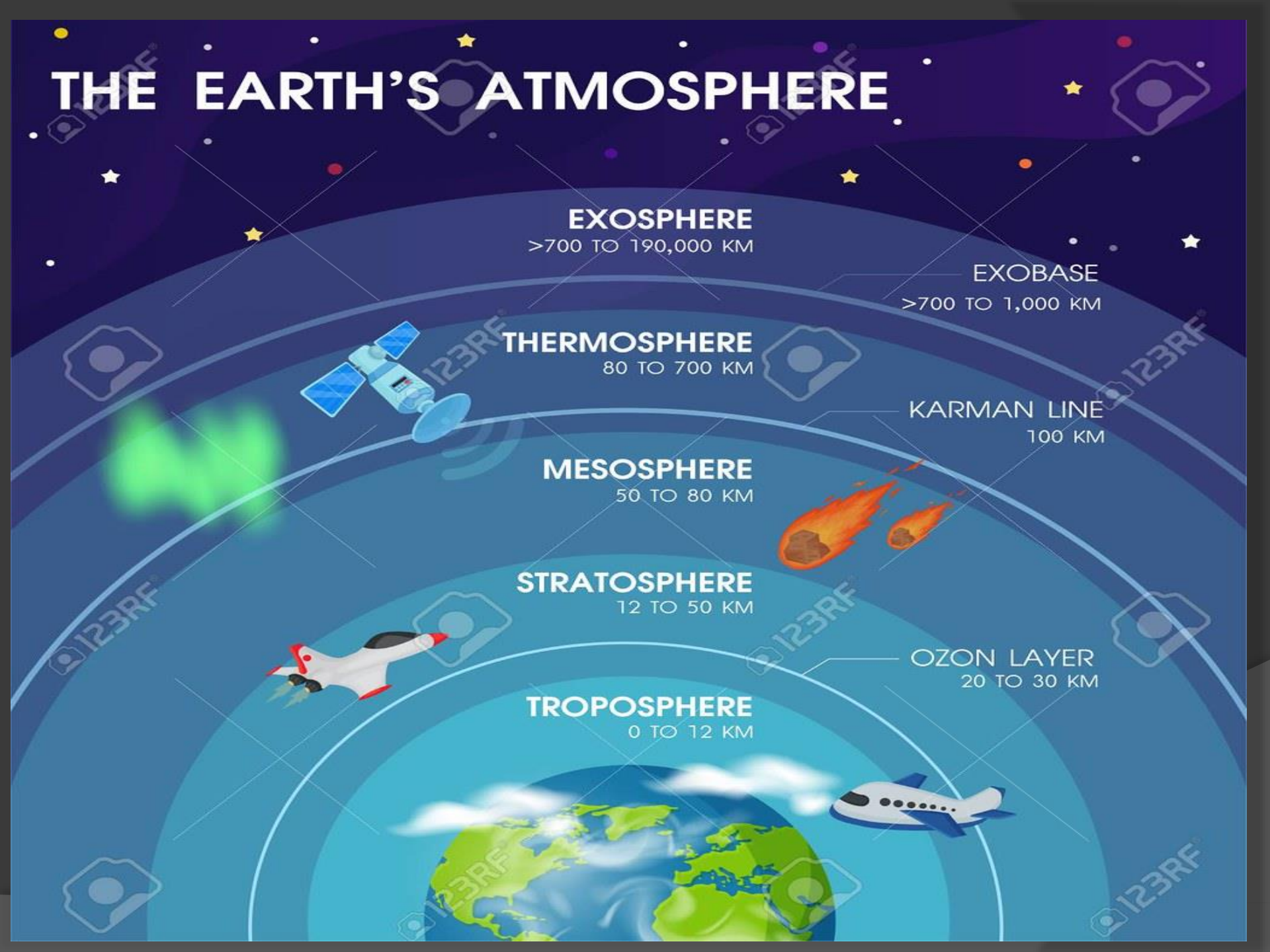
12 TO 50 KM

OZON LAYER

20 TO 30 KM

TROPOSPHERE

0 TO 12 KM



Normal components of our atmosphere

- ❑ **Nitrogen - 78.1%**
- ❑ **Oxygen - 20.9%**
- ❑ **Carbon dioxide - 0.03%**
- ❑ **Everything else - 0.07%**
 - noble gases (krypton, xenon, argon, helium)
 - methane
 - sulfur dioxide

What is air Pollution?

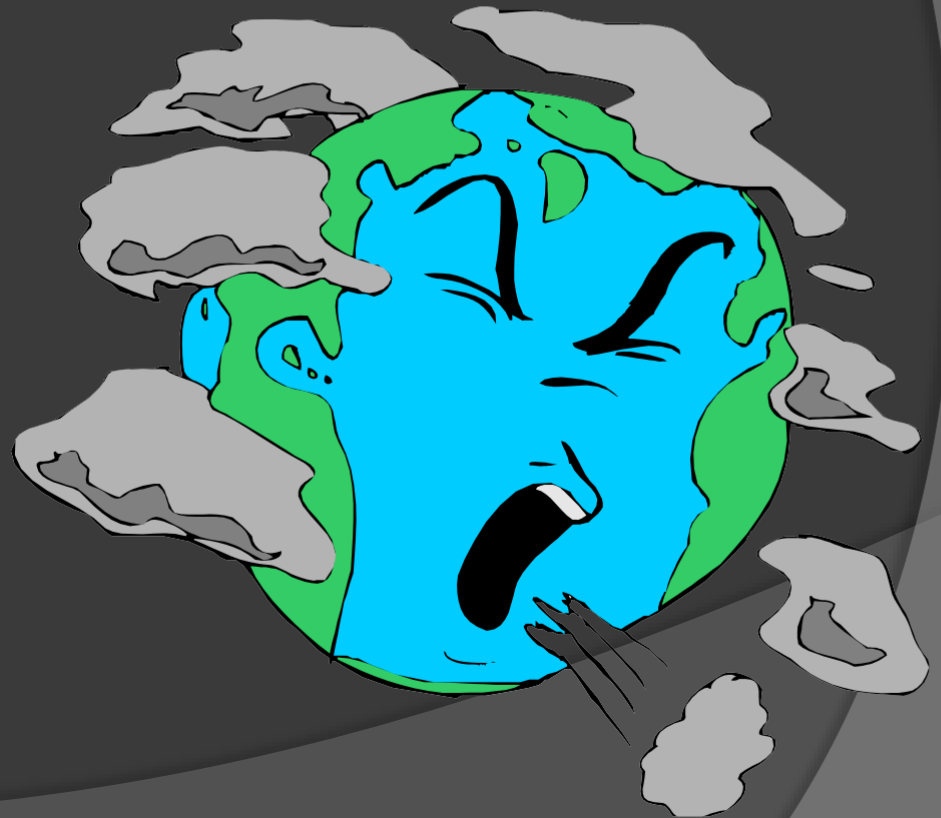
Air pollution is therefore defined as the presence of solid, liquid, or gaseous substance (including noise) present in the atmosphere in such concentration that may or tend to be injurious to human beings or other living creature or plants or property.



Cont....

- ④ may be defined as any atmospheric condition in which certain substances are present in such concentrations that they can produce undesirable effects on man and his environment.

- Any visible or invisible particle or gas found in the air that is not part of the original, normal composition.



Causes Of Air Pollution

- Natural

Air pollution may be caused by natural events like Volcanoes which release huge amount of ash, dust and other gases in the atmosphere or by forest fires. These tends to remain in the atmosphere for a short time.

- Man made

Air pollution may be caused by anthropogenic activities such as burning of fossil fuel, Industrial emission, Urbanization etc. The pollutant released may remain in the air for long duration.

What is air Pollutants?

- Air pollutants may be either emitted into the atmosphere or formed within atmosphere itself. Pollutants are of two types :

1. Primary air pollutants

Sources such as factory chimney or exhaust pipe.

2. Secondary air pollutants

Eg. ozone.

1.Primary Air Pollutants

The atmospheric air may contain hundreds of air pollutants from the natural or the manmade sources. All the pollutants which are emitted directly from identifiable sources are called Primary Pollutants.

The important primary air pollutants are:

1. Oxides of Sulphur mainly SO_2
2. Oxides of carbon mainly CO and CO_2
3. Oxides of Nitrogen NO , NO_2 , NO_3
4. Volatile Organic Compounds mostly Hydrocarbons
5. Suspended Particulate Matter.

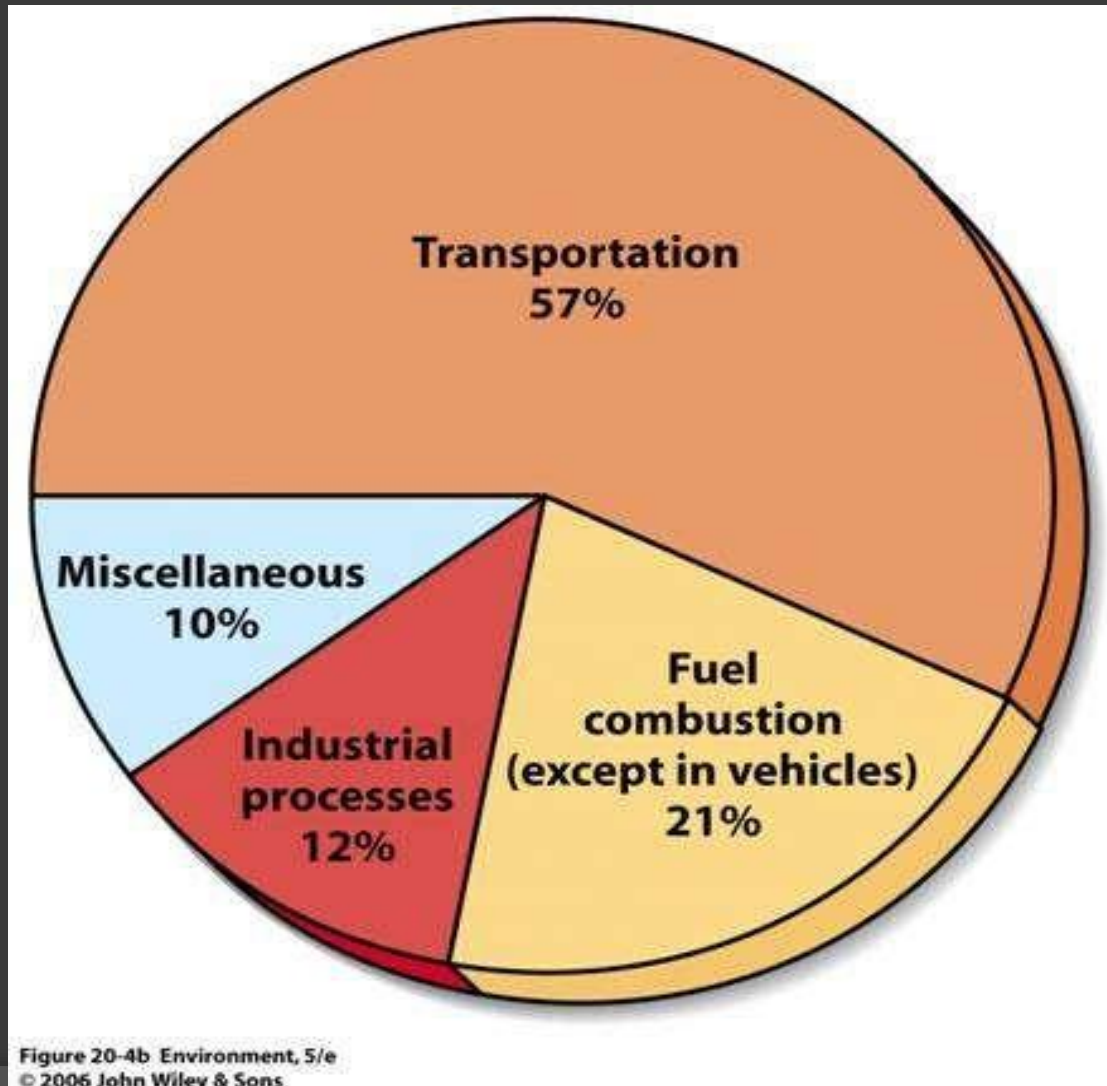
2. Secondary Air Pollutants

These are produced in the air by the interaction among two or more primary pollutants or by reaction with normal atmospheric constituents.

Important secondary pollutants are :

1. SMOG
2. Ozone (O_3)
3. Peroxy Acyl Nitrate (PAN)

Sources of Primary Air Pollutants



The main sources of air pollution are:

(a)AUTOMOBILES:

- ⦿ Motor vehicles are a **major source** of air pollution throughout the urban areas.
- ⦿ They emit hydrocarbons, carbon monoxide, lead, nitrogen oxides and particulate matter.
- ⦿ In strong sunlight, certain of these hydrocarbons and oxides of nitrogen may be converted in the atmosphere into "photochemical" pollutants of oxidizing nature.

The main sources of air pollution are:

- In addition, diesel engines, when misused or badly adjusted are capable of emitting black smoke and malodorous fumes.



The main sources of air pollution are:

(b) INDUSTRIES:

- ⦿ Combustion of fuel to generate heat and power produces smoke, sulphur dioxide, nitrogen oxides and fly ash.
- ⦿ Petrochemical industries generate hydrogen fluoride, hydrochloric acid and organic halides.

The main sources of air pollution are:

- Many industries discharge carbon monoxide, carbon dioxide, ozone, hydrogen sulphide and sulphur dioxide.
- Industries discharge their wastes from high chimneys at high temperature and high speed.

The main sources of air pollution are:

(c) DOMESTIC SOURCES :

- ⦿ Domestic combustion of coal, wood or oil is a major source of smoke, dust, sulphur dioxide and nitrogen oxides.

(d) tobacco smoke:

- ⦿ The most direct and important source of air pollution affecting the health of many people .
- ⦿ Even those who do not smoke may inhale the smoke produced by others ("passive smoking").

The main sources of air pollution are:

(e) MISCELLANEOUS:

- ⦿ These comprise burning refuse, incinerators, pesticide spraying, natural sources (e.g., wind borne dust, fungi, molds, bacteria) and nuclear energy programmes.

All these contribute to air pollution.

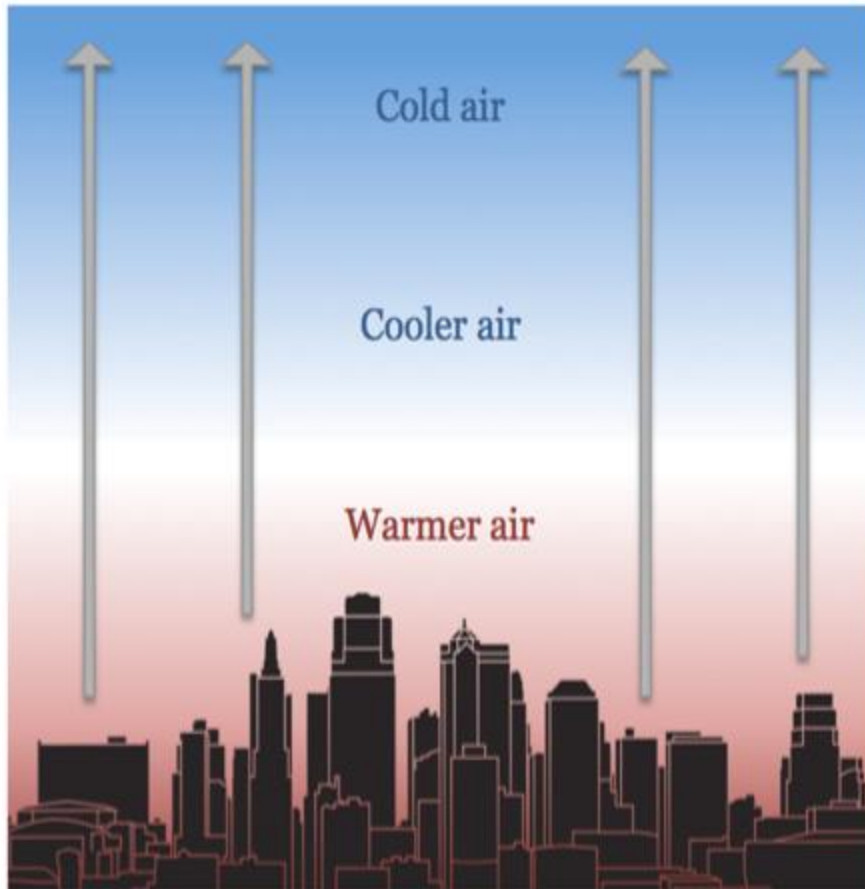
NOTE...

- ⦿ Although the Earth's atmosphere extends to several layers above the surface, it is only the first 30 km that hold the major portion of the atmospheric gases.
- ⦿ Man is most directly concerned with only the 8-10 km of the atmosphere.
- ⦿ The level of atmospheric pollution at any one time depends upon meteorological factors, e.g., topography, air movement and climate.
- ⦿ Winds help in the dispersal and dilution of pollutants.
- ⦿ If the topography is dominated by mountains (or tall buildings) the winds become weak and calm, and pollutants tend to concentrate in the breathing zone.

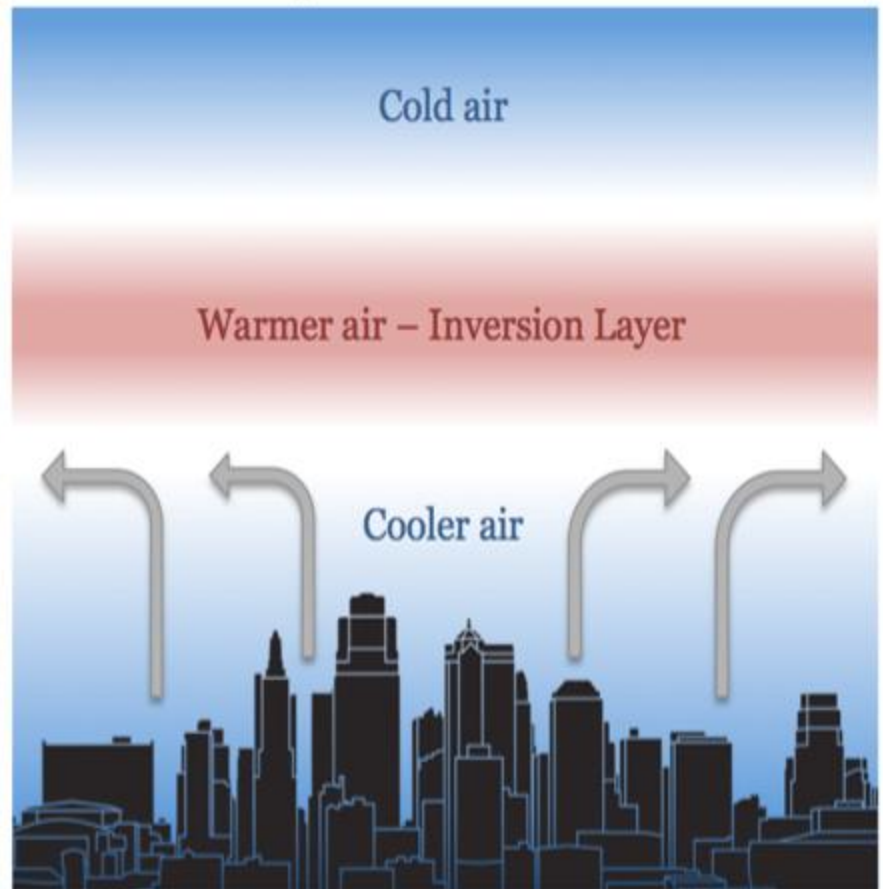
NOTE...

- The vertical diffusion of pollutants depends upon the temperature gradient.
- When there is a rapid cooling of lower layers of air (temperature inversion), there is little vertical motion and the pollutants and water vapours remain trapped at the lower levels and the result is "smog".
- The "temperature inversion "which is more frequent in the winter months than in spring or summer, is a threat to human health.

Normal Conditions



Temperature Inversion



5 Major Pollutants

1. Carbon
Monoxide

2. Sulphur
Dioxide

3. Nitrogen
Dioxide

(1) Carbon monoxide

- It is colorless , odorless gas , a product of incomplete combustion of carbon containing materials, such as in in automobiles, industrial process, heating facilities and incinerators
- Some widespread natural non biological and biological sources have also been identified.
- Concentrations in urban areas depend on weather and traffic density.

Effect of CO pollution on the health:

- It causes harmful effect by reducing oxygen delivery to body organ , in extremely high level it can cause death.
- CO's affinity for Hb is 240–270 times greater than oxygen and Fetal Hb has higher affinity for CO , so it competes with O₂ to bind (irreversibly) with haemoglobin.

Effect of CO pollution on the health:

- By this exposure to it reduce the oxygen-carrying capacity of the blood to the heart, brain and other organs .
- deprives body of O₂ causing headaches, fatigue, MI and impaired vision.

Maximum permissible limit of CO is 10 mg/m³.

(2) Sulphur dioxide

- It is one of the several forms in which sulphur exists in air.
- The others include H_2S , H_2SO_4 and sulphate salts.
- Sulphur dioxide results from the combustion of sulphur containing fossil fuel, and when coal and fuel oil are burned.
- Domestic fires can also produce emissions containing sulphur dioxide.
- Acid aerosol - sulphuric acid (H_2SO_4) is a strong acid that is formed from the reaction of sulphur trioxide gas (SO_3) with water.

Maximum permissible limit is 50 microgram/ m^3 in residential areas.

(3)Nitrogen Dioxide

- Nitric oxide (NO) is produced by combination of Atmospheric nitrogen with oxygen at high temperature..
- Nitrogen dioxide (NO₂), which has greater health effects, is a **secondary pollutant created by the oxidation of NO under conditions of sunlight**, or may be formed directly by higher temperature
- combustion in power plants or indoors from gas stoves.

(3) Nitrogen Dioxide

- reddish, brown gas present in car exhaust and power plants.
- ⦿ Levels of exposure to nitrogen dioxide that should not be exceeded (WHO guideline levels) are respectively 400 $\mu\text{g}/\text{m}^3$ (0.21 parts per million (ppm) for one hour and 150 $\mu\text{g}/\text{m}^3$ (0.08 ppm) for 24 hours (WHO, 1987a).
- ⦿ Maximum permissible limit is 40 microgram/ m^3 .

(4) Particulate matter :

- represents a complex mixture of organic and inorganic substances.
- Mass and composition tend to divide into two principal groups :
- coarse particles larger than $2.5\text{ }\mu\text{m}$ in aerodynamic diameter, and fine particles smaller than $2.5\text{ }\mu\text{m}$ in aerodynamic diameter.
- The smaller particles contain the secondarily formed aerosols ,combustion particles and recompensed organic and metal vapours.
- The large particles usually contain earth's crustal material and fugitive dust from roads and industries.

(4) Particulate matter :

The particulate matter in air may occurs largely in solid form as particles of dust, smoke, fume etc and also in liquid form as mist (Aerosols consisting of liquid droplets eg. H_2SO_4) and fog (Aerosols consisting of water droplets). Aerosols is a suspension of fine solid particles or liquid droplets in air.

As per national ambient air quality standard of India max. annual concentration of PM₁₀ and PM_{2.5} are 60 microgram/m³ and 40 microgram/m³ respectively.

- Particulate matter of respirable size may be emitted from a number of sources, some of them natural (e.g. dust storms) and many others that are more widespread and more important (e.g., power plants and industrial processes, domestic coal burning, industrial incinerators).
- Bigger particles >10 Micrometre are trapped by hairs and sticky mucus in the lining of respiratory track.

(5) Hydrocarbons

They are group of compounds consisting of carbon and hydrogen atom.

They are evaporated into the atmosphere from automobiles, and also from refineries at high temperature. They cause irritation pf eyes and respiratory tract.

They are of two types:

1. Aliphatic hydrocarbons:- It includes alkenes, alkynes.
Alkenes are highly reactive
2. Aromatic hydrocarbons:- Many of them are carcinogenic. Eg. Benzorthopyrene

Maximum permissible limit for Benzene and benzopyrene are 5 microgram/m³ and 1 microgram/m³

These 5 together form.....

SMOG

Secondary Air Pollutants

1. SMOG
2. OZONE

SMOG

- Smog is made up of mixture of smoke and fog.
- Smog is caused by the interaction of some hydrocarbon and oxidants like NO_2 and SO_2 .
- There are two types : reducing smog characterized by sulphur dioxide and particulate.
- And photochemical smog characterized by ozone and other oxidants.

- 1st smog related deaths were in London in 1873; death of 500 people; can you imagine how much worse the atmosphere is now?!

- Limits visibility

- Decreases UV radiation

- Yellow/black color over cities

- Causes respiratory problems and bronchial related deaths

Ozone

Ozone is a gas found in traces in the air near the ground surface and in concentrated thick layer at varying heights in the troposphere from 16 Km to about 40 Km at different latitudes. It is an toxic air pollutant gas.

This gas is extremely rare in the atmosphere, representing just three out of every 10 million molecules.

- Ozone is highly corrosive and toxic and is used as a disinfectant.
- Ozone gas can be created or destroyed by the sun's UltraViolet rays.
- Ozone levels are reported in Dobson Units (DU) & 300 DU is an average value.
- The Ozone found at Stratosphere is a Good Ozone as it absorb the harmful UV rays of Sun
- When an Oxygen molecule absorbs a Photon of Light with a wavelength shorter than 200 nanometers (1 Billion of meter) the energy splits the molecule into two Oxygen atoms. One of these atoms can react with another oxygen molecule to form an Ozone Molecule.



PAN

- Peroxyacetyl nitrate is a secondary pollutant present in photochemical smog. It is thermally unstable and decomposes into peroxyethanoyl radicals and nitrogen dioxide gas. It is a lachrymatory substance.
- Peroxyacetyl nitrate, or PAN, is an oxidant that is more stable than ozone. Hence, it is more capable of long-range transport than ozone. It serves as a carrier for oxides of nitrogen (NO_x) into rural regions and causes ozone formation in the global troposphere.
- PANs are both toxic and irritating, as they dissolve more readily in water than ozone. They are lachrymators, causing eye irritation at concentrations of only a few parts per billion. At higher concentrations they cause extensive damage to vegetation. Both PANs and their chlorinated derivatives are said to be mutagenic, as they can be a factor causing skin cancer.

Volatile organic compounds:

- ◎ VOCs are divided into the separate categories of methane (CH_4) and non-methane (NMVOCs)
- ◎ Methane is an extremely efficient greenhouse gas which contributes to enhanced global warming
- ◎ NMVOCs, the aromatic compounds benzene, toluene and xylene are suspected carcinogens and may lead to leukaemia through prolonged exposure .

Toxic metals:

Lead

Cadmium

Copper

- ⦿ Those metals are produced from different factories and can have immediate health effects and long term effects on human health

Others...

- **Chlorofluorocarbons: (CFCs) –**